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Application Note

Advanced morphometric classification with Classika

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Abstract

Motivation: Morphologika data offers promise in the field of evolutionary biology as a data type that helps researchers uncover phylogenetic relationships with physical measurements of organisms (Ammresh *et al.*). Classification of species based on Morphologika data using machine learning methods has the potential to aid researchers in scientific inquiry by offering insights into evolutionary history using 2D and 3D data that may otherwise be difficult to analyze. We present Classika, a locally hosted web app that takes labeled Morphologika files, calculates metrics, and evaluates eleven machine learning models on their performance in classifying those data. We believe developing a strong foundation for the performance of a machine learning approach to species classification based on Morphologika data has the potential to produce novel scientific methods without the need for labeled data, expediting discovery of evolutionary relationships.

Availability: This report, along with the entire Classika app, are available at https://github.com/otodreas/Classika

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1 Introduction

Classika (morphometric classification program for Morphologika files) is a local host web app designed to take labeled Morphologika files as input and learn to label those data with a set of machine learning algorithms. It relies on the development of a multi-model machine learning-based program to classify Morphologika data based on 2D and 3D features. This program is what Classika is built around. The core logic of Classika reads landmark coordinate data in Morphologika format and derives a set of shape variables from those data, building a feature matrix. This feature matrix is what multiple machine learning models are trained and evaluated on. Since the data are labeled, the performance of each model can be evaluated, which is the value proposition of Classika. It is a tool that allows the user to evaluate the classification performance of many models on labeled Morphologika data.

Classika shows the potential of multi-model approaches in classifying morphological data measurements of individuals into species. Previous studies have shown that overreliance on PCA dimensionality reduction can impact performance, which is why Classika opts for an engineered feature matrix rather than principal components in its core logic (Mohseni & Elhaik, 2024).

The novelty of Classika is in its web app interface that allows users without programming experience to examine the performance of the many models built into Classika. It may prove useful to researchers looking to validate existing classifications, assess dataset quality, and explore the differences between classifiers on their data. For instance, a dataset that Classika struggles to classify may raise warnings about the quality of the data or the accuracy of the existing labels.

2 Methods

Classika 1.1 is built entirely in Python 3.11 and up as of time of release. The frontend of the interface is based on Streamlit 1.55, which allows Classika to function as a locally hosted web app. One or more Morphologika files with labels may be passed to the program in the user interface. In addition, both global and model-specific parameters are made accessible to the user, making it possible to tweak the program to maximize performance on a given dataset.

Following data importing and parameter setting, data are validated before the analysis is run. After inputs are read, shape descriptors such as pairwise Euclidean distances between landmarks, centroid-based shape features, and angles between landmark triplets are calculated from the Morphologika files and concatenated into a feature matrix used in classification.

Eleven models are trained to classify the samples based on the calculated features: XGBoost, LightGBM, CatBoost, Random Forest, Extra Trees, Gradient Boosting (sklearn), Logistic Regression, Linear Discriminant Analysis (LDA), Support Vector Machine (SVM), neural network (MLP), and k-nearest neighbor. In addition, ensemble strategies are evaluated. Model performance is then evaluated using k-fold cross-validation so that each sample is predicted once, controlling for favorable or unfavorable training set partitioning, according to the methods of Mohseni & Elhaik (2024).

The user may opt to build a weighted voting ensemble in the process. In the evaluation step, models are scored on their accuracy, and the top models are selected (the number of top models selected can be set by the user). Weights are assigned based on their accuracy, and models produce probability distributions of classes. The product of the weights and probability distributions determine the final classification decisions of the voting ensemble.

The results produced by Classika include confusion matrices for the best performing model and an Excel spreadsheet with multiple sheets that outline parameters used, model performance, and ensemble tuning metrics. A json file summarizing the weighted model voting is also provided if requested by the user.

A test dataset along with the expected Classika output is provided in a Classika installation. The dataset contains skull and bone morphology data from different primates including *Papio cynocephalus* (yellow baboon) and *Macaca mulatta* (Rhesus macaque).

3 Features

Classika validates input files, meaning that they must be in plain text format and have Morphologika-style headers: "[individuals]," "[landmarks]," "[dimensions]," "[names]," and "[rawpoints]." Furthermore, Classika expects individuals in the "[names]" section to be labeled following binomial nomenclature, with genus and species name being separated by a space or an underscore.

Program runtime is significantly affected by the number of CV folds used to fragment the data during training and testing partitioning. A higher number of folds, while it can improve performance, may slow down the performance of the program from minutes to hours. To show the user that the program is running, the output log is streamed to the interface, allowing the user to see which step the classification is in.

Since outputs are offered for download, it is imperative that they are easily understood. For this reason, there are only a small number of outputs: a run log, a result spreadsheet, one confusion matrix per dataset, and a folder for miscellaneous outputs. The run log is identical to the text streamed to the interface during analysis. The result spreadsheet contains information on model performance and parameters used, essential for making the workflow reproducible. The confusion matrices allow the user to at a glance determine the performance of the best model.

Classika's web app interface means that users without programming experience can run many machine learning analyses on Morphologika data. Nonetheless, a manual installation of Python, Classika, and Classika's dependencies is required of the user. However, instructions are detailed in the README of the app. The entire program is offered for download in releases, giving the user access to all versions of Classika, improving reproducibility.

4 Discussion

Classika addresses a specific need in the fields of evolutionary biology and phylogenetics: it allows scientists without programming experience to examine the performance of a range of machine learning classification methods on a labeled Morphologika dataset. This is useful for researchers looking to assess the accuracy of labels in their dataset, the quality of landmark data in their dataset, or for those looking to find optimal machine learning algorithms to apply to labeling tasks for similar datasets. While Classika currently only supports model validation on labeled data, there is much to learn from a Classika run. For instance, if the labels produced by the best model mostly disagree with the labels in the dataset, there may be cause for concern regarding the landmark coordinate signal or label quality.

Future Classika releases may work to expand on the functionality of the app. Supporting classification of unlabeled data offers a major area for improvement but certainly requires extensive work. Simplifying the installation experience and estimating remaining time in the analysis are further areas for improvement. Nonetheless, Classika is at present a simple to use app that has the potential to elevate the quality of scientific inquiry across the fields of evolutionary biology and phylogenetics.

Acknowledgements and conflict of interest

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References

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